ST. ANDREW'S JUNIOR COLLEGE
2008 JC2 Preliminary Examinations

H2 BIOLOGY 9747/1

Paper 1 Multiple Choice

Friday 12 September 2008 1 hour

Additional materials: Multiple Choice Answer Sheet
Soft clean eraser (not supplied)
Soft pencil (type B or HB is recommended)

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, civics group and index number on the optical answer sheet in the spaces provided.

There are **40** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the one you consider correct and record your choice in **soft pencil** on the separate multiple choice answer sheet.

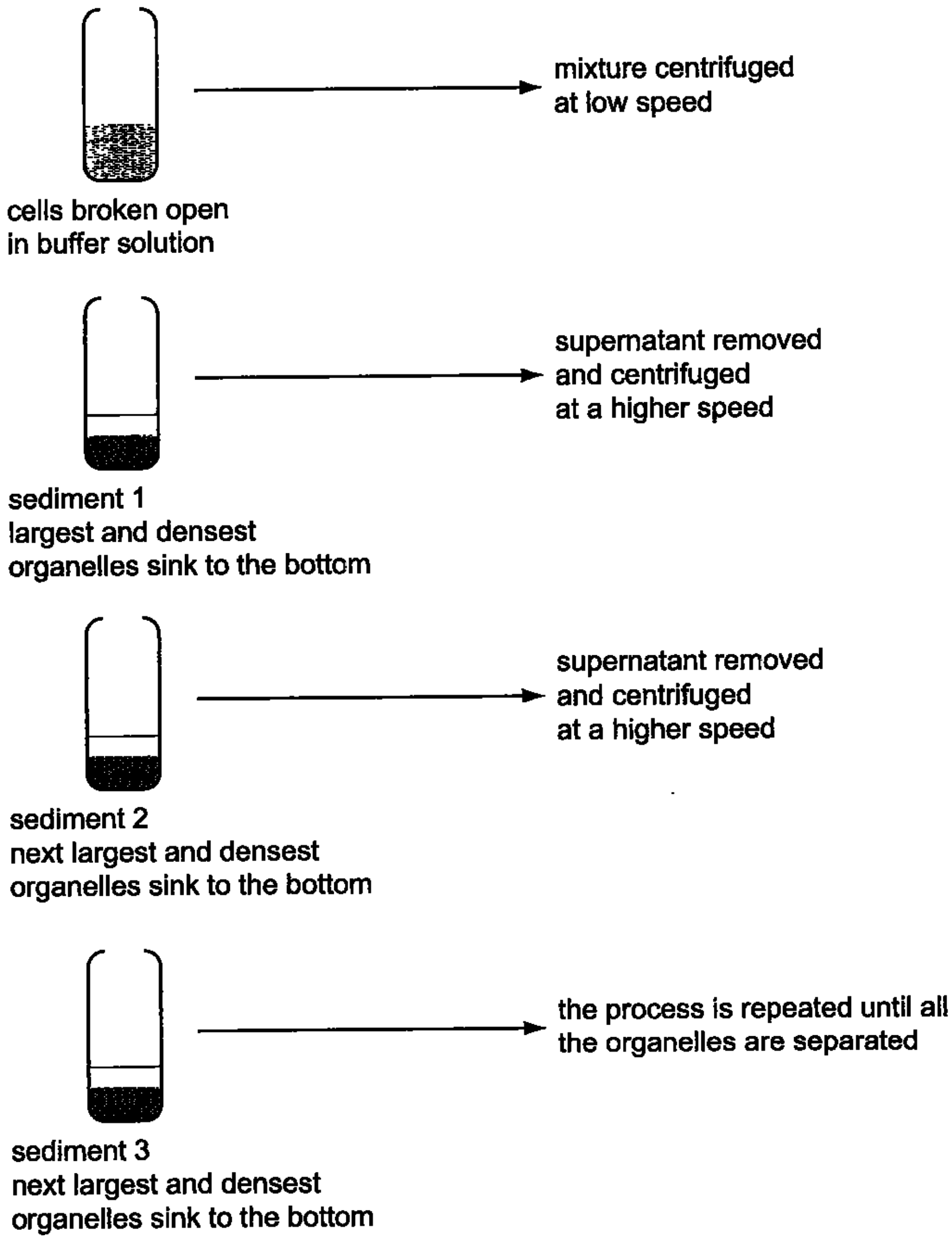
INFORMATION TO CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for wrong answer.
Any rough working should be done in this booklet.

At the end of the examination, submit the optical answer sheet only.

This document consists of **17** printed pages.

1 Fractionation is a process used to separate cell components according to their size and density. The diagram shows the main steps in fractionation of a plant cell.



DCPIP and buffer solution were added to each sediment and the mixtures left in the light for fifteen minutes. Sediment 2 caused the oxidised DCPIP to be reduced.

Which organelles are present in sediment 2?

A chloroplasts
B mitochondria
C nuclei
D ribosomes

2 The following structures can be seen inside cells using an electron microscope

1 chloroplast
2 endoplasmic reticulum
3 Golgi body
4 Lysosome
5 Mitochondrion
6 Nucleolus
7 Nucleus
8 ribosome

Which list contains four membrane-bound organelles?

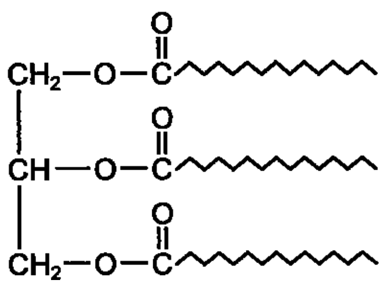
A	1	3	6	7
B	1	4	5	7
C	2	3	4	8
D	2	5	6	8

3 Which of the following(s) is/are true about cellulose and collagen?

1. Both serve as structural components as they have high tensile strength
2. The high tensile strength is a result of interchain hydrogen bonds
3. Both show quaternary level of organization

A 1 only
B 1 and 2 only
C 1, 2 and 3 only
D 2 only

4 The diagram shows a generalized triglyceride molecule.



In the triglyceride above, identify the molecules attached to the glycerol and name the types of bonds.

A three amino acids by ester bonds
B three amino acids by peptide bonds
C three fatty acids by ester bonds
D three fatty acids by peptide bonds

5 The correct shape of a globular protein is maintained by several types of bond.

Bond types
1 disulphide
2 hydrogen
3 ionic
4 peptide

Which bond types are involved in maintaining the different levels of structural complexity of a globular protein?

	Level of structural complexity		
	Primary	Secondary	Tertiary
A	4	3	1,2,3
B	2	1	2,3,4
C	2	4	2,3,4
D	4	2	1,2,3

6 Succinic dehydrogenase is the enzyme that catalyzes the oxidation of succinic acid during cellular respiration. If malonic acid is added to the system, the rate of reaction is reduced. An increase in substrate concentration will, however, increase the rate of reaction again.

From the above information, what can you conclude about malonic acid?

A It binds to the allosteric site of the enzyme
B It has a permanent attachment to the active site of the enzyme
C It has a similar molecular configuration to that of succinic acid
D It increases the pH of the system

7 The list shows the stages in the cellular synthesis of proteins.

1 movement of mRNA from nucleus to cytoplasm
2 linking of adjacent amino acid molecules
3 transcription of mRNA from a DNA template
4 formation of the polypeptide chain
5 attachment of the mRNA strand to a ribosome

Which sequence is correct?

A $1 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 4$
B $1 \rightarrow 5 \rightarrow 3 \rightarrow 4 \rightarrow 2$
C $3 \rightarrow 1 \rightarrow 5 \rightarrow 2 \rightarrow 4$
D $3 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 5$

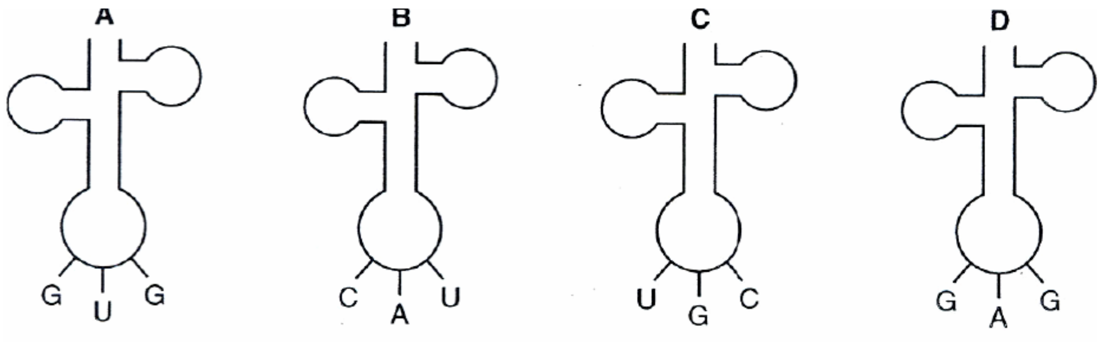
8 Part of the amino acid sequences in normal and sickle cell haemoglobin are shown.

Normal haemoglobin	Sickle cell haemoglobin
thr – pro – glu – glu	thr – pro – val – glu

mRNA codons for these amino acids are :

Name of amino acid	mRNA codons	
Glutamine (glu)	GAA	GAG
Threonine (thr)	ACU	ACC
Proline (pro)	CCU	CCC
Valine (val)	GUA	GUG

Which transfer RNA molecule is involved in the formation of this part of the sickle cell haemoglobin?



9 In most organisms, all possible triplets of bases in DNA code for an amino acid, except for the coding strand 'stop" triplets ATT, ATC, ACT. The coding strand is complementary to mRNA.

In the prokaryote, *Methanosarcina barkeri*, the coding strand triplet ATC codes for an unusual amino acid, pyrrolysine.

The non-coding strand of a section of DNA in *M.barkeri* has the following sequence of bases.

TTTTTATTGTATTACTAGTGTTAATGA

How many amino acids can be made into a peptide by *M.barkeri* from the **coding** strand of this region of DNA?

A 5
B 6
C 7
D 9

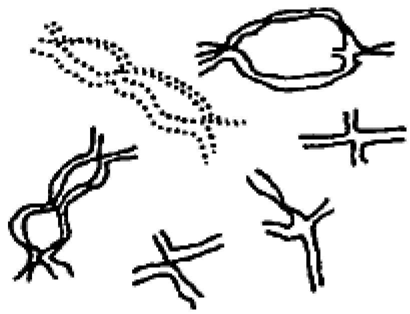
10 Mitosis is a form of cell division, used in growth, repair and asexual reproduction. What is the particular importance of mitosis to living organisms?

A It is a rapid process
B It results in genetically identical daughter cells
C Each division produces a large number of cells
D It reduces the number of chromosomes in each cell

11 If X units of DNA are present in the nucleus of a cell, which has just divided, what is the relative amount present in this cell during anaphase of the next mitosis?

A X/4
B X/2
C X
D 2X

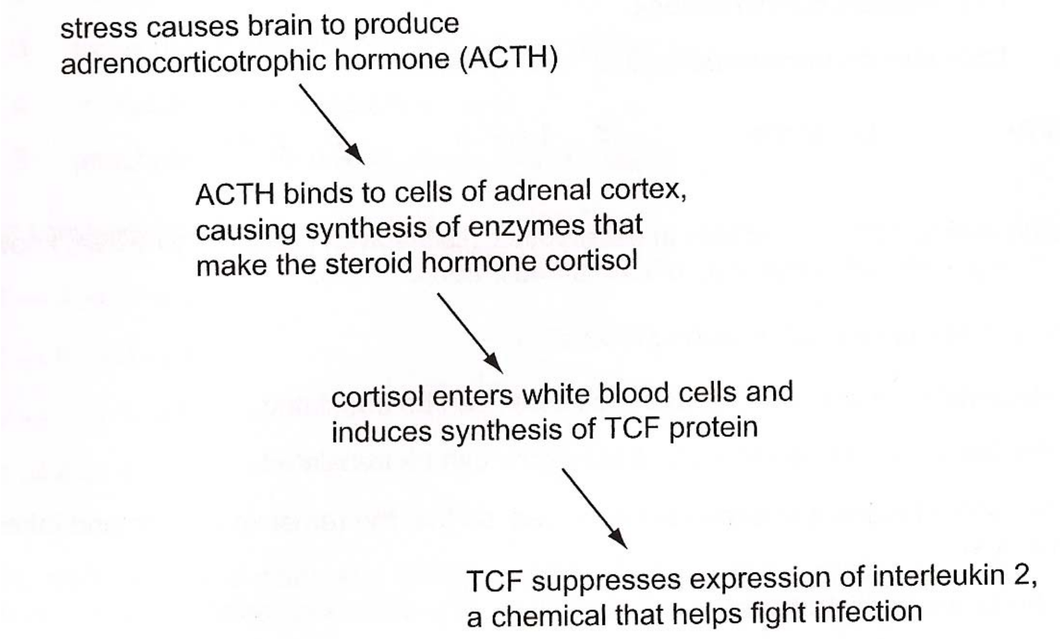
12 The figure below shows the chromosomes from one cell in the ovary of an animal. The cells are in late prophase of its first division.



What is the number of chromosomes present in the cells at the start of the second division and in the liver cells of the same animal?

	Start of second division	*Liver cells*
A	3	6
B	3	12
C	6	12
D	12	24

13 When a person undergoes a stressful experience, their immune system can be depressed and they become more susceptible to infection. Some of the elements involved in this chain of events are shown in the diagram below.



Which combination correctly shows the genes that have transcription-enhancing factors bound to their control elements during the above sequence of events?

	Gene for ACTH	Gene for TCF	Gene for interleukin 2
A	√	X	X
B	X	√	√
C	√	√	X
D	X	X	√

14 Four different genes are regulated in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing
Gene 2 is part of a group of structural genes controlled by the same regulatory sequences
Gene 3 is in some circumstances subject to methylation
Gene 4 codes for a repressor protein which acts at an operator site close by

Which role of the table correctly identifies which genes are prokaryotic and which are eukaryotic?

	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

15 Which of the following is not a requirement for the mechanism of RNA splicing?

A branchpoint sequence
B small nuclear ribonucleoproteins
C 5' and 3' splice site
D 5' modified guanosine cap

16 Which feature occurs in the life cycle of the influenza virus?

A Host cell DNA is destroyed by lytic enzymes
B Viral genome is integrated into the host genome
C Viral DNA acts as a template for DNA synthesis
D Viruses enter the host cell by receptor-mediated endocytosis

17 Some events that take place during generalized transduction are listed below.

1 Bacterial host DNA is fragmented
2 Bacterial DNA may be packaged in a phage capsid
3 Recombination between donor bacterial DNA and recipient bacterial DNA
4 Phage infects a bacterial cell
5 Phage DNA and proteins are made

Which sequence of events is correct?

First -----------------→ Last
A 4,1,3,5,2
B 4,1,5,2,3
C 4,3,1,5,2
D 4,5,1,3,2

18 If glucose and lactose are both absent from the growth medium, and the *lac* structural genes are expressed efficiently, what would you suspect about the condition of the *E.coli* cell?

1 The cell has a mutation in the operator of the *lac* operon.
2 The cell has a mutation in the *lacI* gene.
3 The cell has a mutation in the promoter of the *lac* structural gene.

A **1** only
B **2** only
C **1** and **2**
D **2** and **3**

19 A virus has a base ratio of (A + G) / (U + C) = 1. What type of virus is this?

A a single-stranded DNA virus
B a double-stranded DNA virus
C a single-stranded RNA virus
D a double-stranded RNA virus

20 In fruit flies, one gene controls wing form (normal or vestigial) and one gene controls eye colour (red or normal brown). A fly with normal wings and normal brown eyes is crossed with a fly with vestigial wings and red eyes. All the F1 are normal for both characteristics.

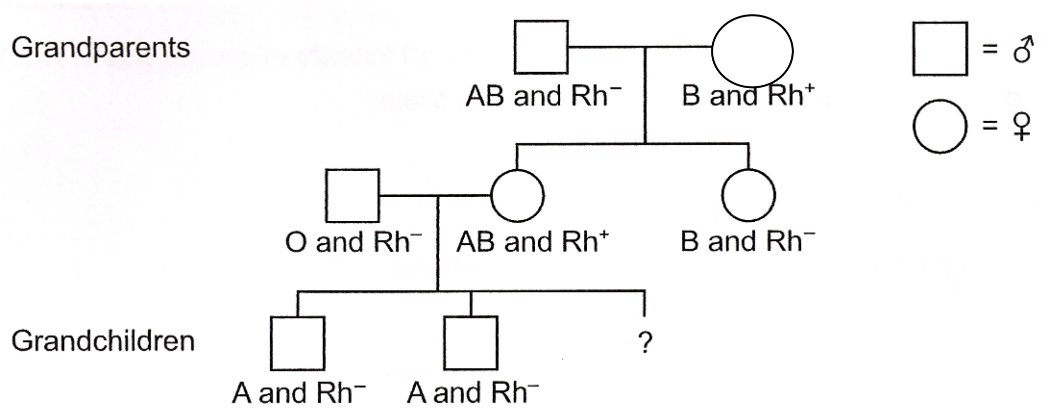
However, when F1 are crossed with each other, the resulting F2 is:

45 normal wing, normal brown eye
17 normal wing, red eye
16 vestigial wing, normal brown eye
5 vestigial wing, red eye
1 normal wing, orange eye

What is the best explanation for the results of this dihybrid cross?

A codominance
B gene mutation
C multiple alleles
D sex linkage

21 The diagram shows a family tree.



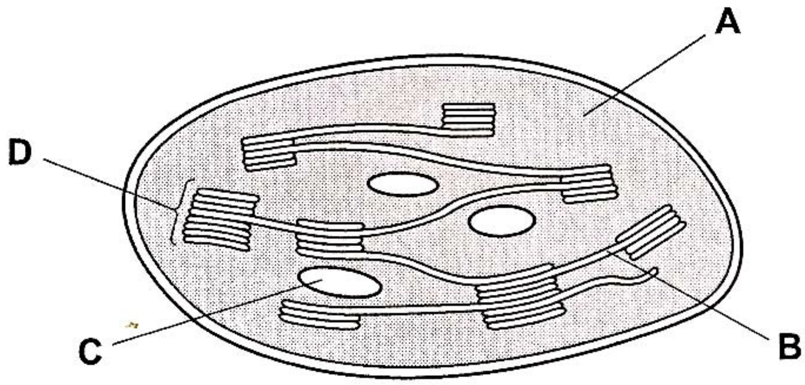
What is the probability that the third grandchild will be a boy with blood group B and Rh positive blood?

A 0.0625 (1 in 16)
B 0.125 (1 in 8)
C 0.25 (1 in 4)
D 0.5 (1 in 2)

22 Which of the following incorrectly describes the features of continuous variation?

A no distinct categories into which individuals can be placed
B quantitative, with overlaps between categories
C polygenic, where inherited individual alleles have an additive effect.
D not affected by the environment

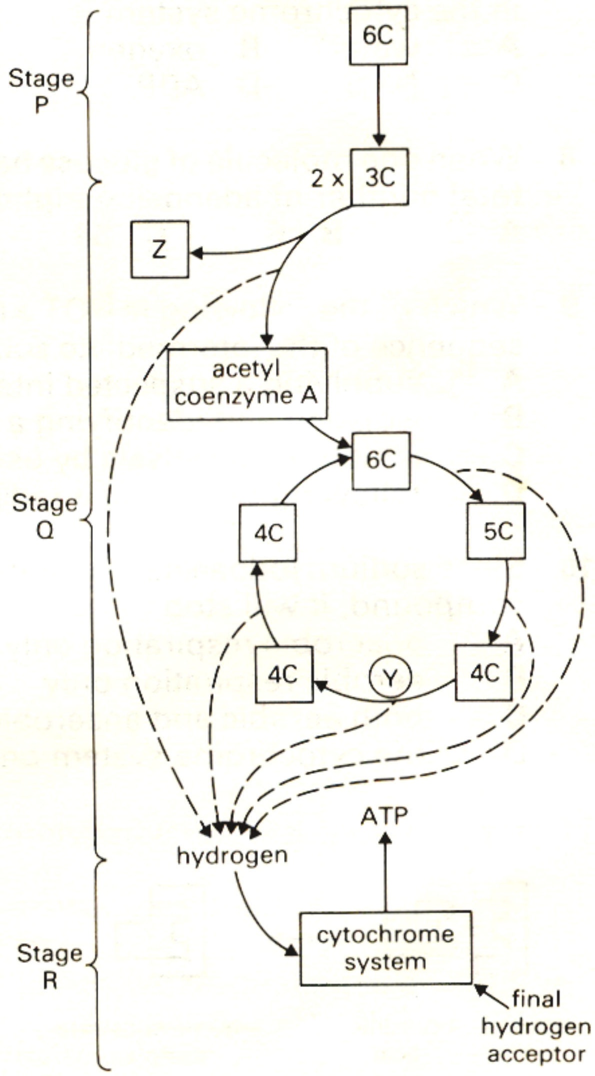
23 The diagram shows a section through a chloroplast. Where would the products of photophosphorylation be used?



24 Which statement about anaerobic respiration is correct?

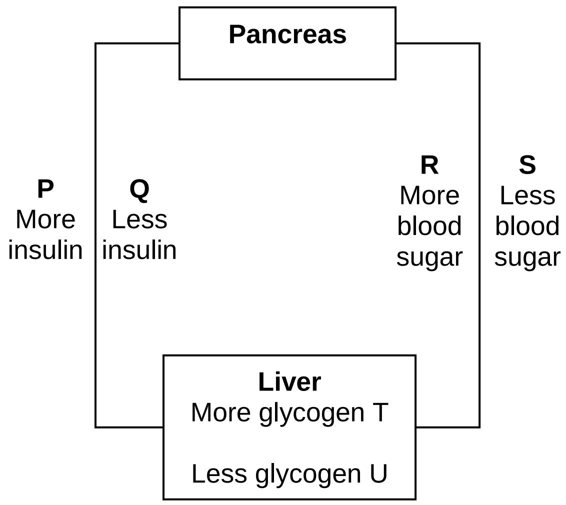
A From one molecule of glucose, ethanol and lactate production yield the same amounts of ATP
B From one molecule of glucose, ethanol production yields less energy than lactate production
C From one molecule of glucose, ethanol production yields more energy than lactate production
D Yeast is able to respire ethanol for the production of ATP

25 Which of the following indicates the correct locations at which stages P, Q and R occur in a cell?



	Cristae	Cytosol	Matrix
A	R	P	Q
B	Q	P	R
C	P	R	Q
D	R	Q	P

26 Which of the following responses below would occur in a human body as a direct result of eating a meal?



A R, P and T
B S, P and T
C R, Q and T
D R, P and U

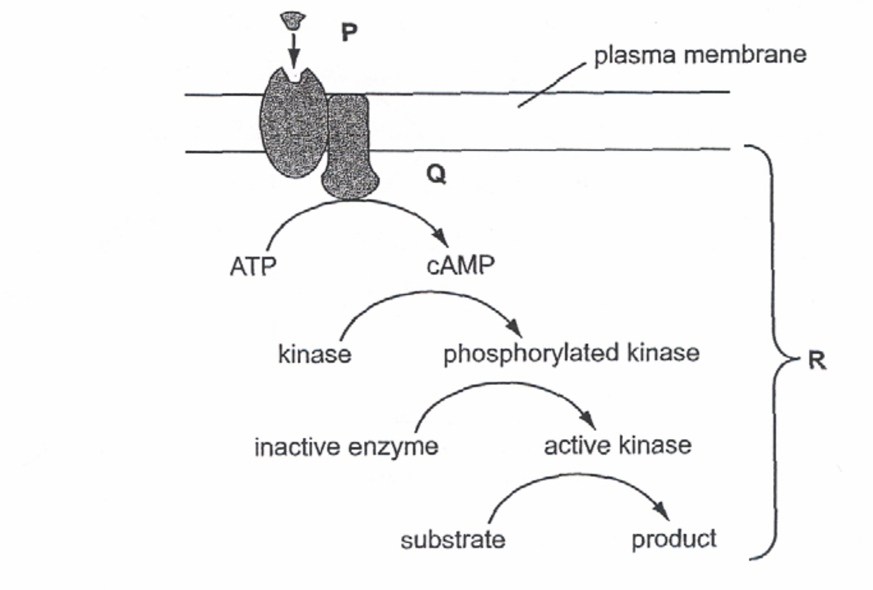
27 A particular type of autoimmune disease causes antibodies to destroy acetylcholine receptors of neurons. What effect will this have on the functioning of the nervous system?

A Presynaptic neurons will be unable to release neurotransmitters into the synaptic clefts.
B Neurons will be unable to propagate action potentials along their axons.
C Depolarized neurons will be unable to re-establish an ionic gradient across their membranes.
D Postsynaptic neurons will be unable to detect signals from presynaptic neurons.

28 The post-synaptic membrane of a nerve may be stimulated by certain neurotransmitters to permit the influx of negative chloride ions into the cell. This process will result in

A membrane depolarization.
B an action potential.
C the production of an IPSP.
D the production of an EPSP.

29 The diagram shows an example of cell signaling.



Which processes are shown on the diagram?

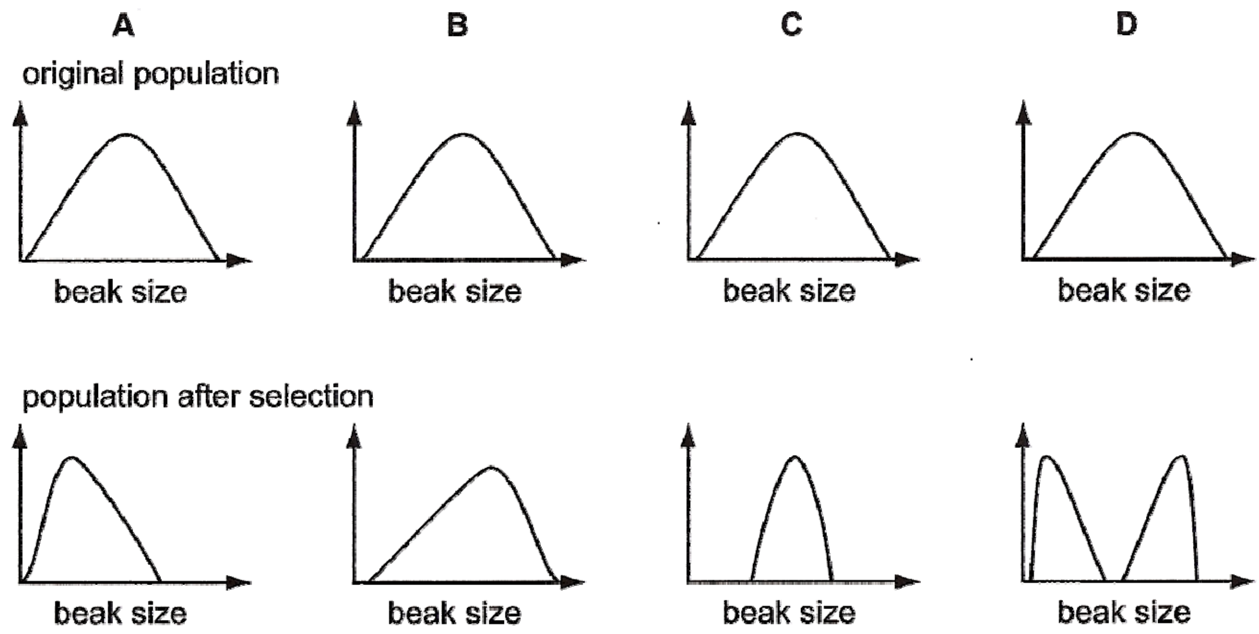
	P	Q	R
A	Amplification	Ligand-receptor interaction	Phosphorylation
B	Ligand-receptor interaction	Amplification	Phosphorylation
C	Transduction	Phosphorylation	Amplification
D	Ligand-receptor interaction	Transduction	Amplification

30 Many signal transduction pathways use second messengers to

A transport a signal through the plasma membrane.
B relay a signal from the outside to the inside of the cell.
C relay a signal from the inside of the membrane throughout the cytoplasm.
D amplify the message by phosphorylating proteins.

31 One species of finch on an island shows variation in beak size. Birds with larger beaks can eat larger seeds. Following a dry period, fewer small seeds are available, with larger seeds and nuts being more plentiful.

Which graph shows the effect of natural selection on beak size of these birds following a dry period?



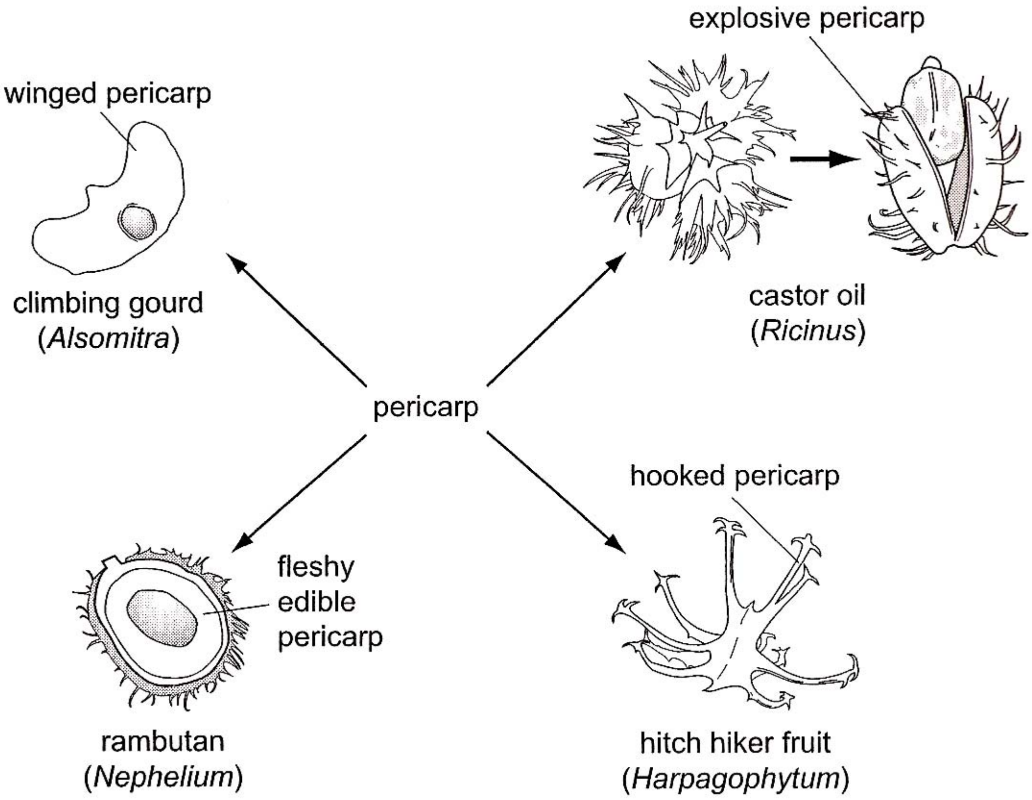
32 Three of the events that occur during speciation are

1 mutation
2 natural selection
3 isolation

The correct order in which these occur is

A 3,2,1
B 2,1,3
C 3,1,2
D 2,3,1

33 The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



What do these examples illustrate?

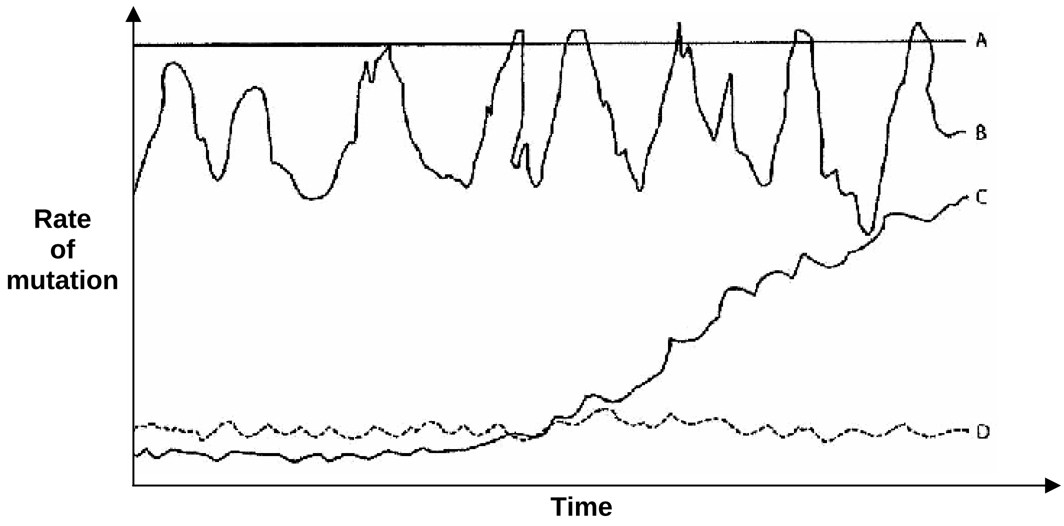
A The adaptive radiation of analogous structures showing convergent evolution

B The adaptive radiation of analogous structures showing divergent evolution

C The adaptive radiation of homologous structures showing convergent evolution

D The adaptive radiation of homologous structures showing divergent evolution

34 Which pattern of mutation rate would be most helpful if one desires to use a gene as a molecular clock to determine evolutionary relatedness of species that are closely related to each other?



35 In which situation would evolution be slowest for an interbreeding population?

	Migration	Selection Pressure	Variation due to mutation
A	high	high	low
B	high	low	high
C	absent	high	high
D	absent	low	low

36 At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60^{0}C.

What explains why the denatured DNA strands anneal with primers and **not** each other?

A The primers are shorter and anneal more easily
B The primers anneal only to the 3' end of the denatured DNA
C The primer concentration exceeds the denatured DNA concentration
D The temperature prevents the denatured DNA from annealing together

37 What is not expected to be an outcome of the Human Genome Project?

A Identification of all genes present in human beings
B Identification of all alleles present in human beings
C Map and sequence the genetic makeup of every human being
D Map and sequence all the genetic material present in the chromosomes of human beings

38 Stem cells are widely used in medical research.

Which property of stem cells makes them particularly useful in this research?

A They can be fused together to form a zygote
B They can divide and eventually give rise to a whole organism
C They can divide and be made to differentiate into various types of cell
D They will continue to divide indefinitely

39 A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

1 Transfer of DNA to nitrocellulose membrane.
2 Restriction digestion of genomic DNA.
3 Cleaved DNA separated using gel electrophoresis.
4 Create radioactive probe.
5 Incubate probe and membrane.

Which is the correct sequence to the above steps?

A $2 \rightarrow 3 \rightarrow 1 \rightarrow 4 \rightarrow 5$
B $2 \rightarrow 3 \rightarrow 1 \rightarrow 5 \rightarrow 4$
C $4 \rightarrow 5 \rightarrow 1 \rightarrow 2 \rightarrow 3$
D $5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1$

40 Each of the statements is concerned with the success or failure of gene therapy as a permanent cure for genetic disorders.

1 Somatic cells cannot pass the modified gene on to any offspring
2 The treatment does not last long as the treated cells die and are replaced
R The treatment does not result in all cells having the therapeutic gene

Which statement(s) explain the limited success of gene therapy as a permanent cure for genetic disorders in human population?

A 1 only
B 1 and 3
C 2 and 3
D 1,2 and 3

END OF PAPER 1